TOPICS ,DATASET USED, CODE APPILED:

Day 7:

Topics Covered: Machine Learning, Supervised Learning, Unsupervised Learning, Reinforcement Learning, AI, Deep Learning, ML Workflow

Dataset Used: Student Performance Dataset (Study Hours vs Pass/Fail)

Code Applied: Data preprocessing, train-test split, model training, prediction using Scikit-Learn.

Day 8:

Topics Covered: Regression, Linear Regression, R^2 Score, MAE, RMSE, Gradient Descent

Dataset Used: Student Study Hours Dataset, House Price Dataset

Code Applied: Linear Regression model building, prediction, evaluation metrics calculation.

Day 9:

Topics Covered: Classification vs Regression, Logistic Regression, Sigmoid Curve, Confusion Matrix, Precision, Recall, F1-Score, ROC Curve, AUC

Dataset Used: Heart Disease Dataset, Diabetes Dataset

Code Applied: Logistic Regression, confusion matrix generation, ROC-AUC analysis, classification prediction.

Day 10:

Topics Covered: Decision Tree, Overfitting, Underfitting, Gini Impurity, Classification vs Regression

Dataset Used: Iris Dataset, Customer Classification Dataset

Code Applied: Decision Tree model training, tree visualization, max_depth tuning, prediction.

Day 11:

Topics Covered: Random Forest, Feature Importance, OOB Score, Voting, n_estimators

Dataset Used: Iris Dataset, Customer Classification Dataset

Code Applied: Random Forest model building, feature importance analysis, ensemble learning techniques.

Day 12:

Topics Covered: Unsupervised Learning, K-Means Clustering, Elbow Method, Distance Metrics

Dataset Used: Iris Dataset, Sample Clustering Dataset

Code Applied: K-Means clustering, cluster visualization, elbow method implementation.

Day 13:

Topics Covered: Neural Networks, Artificial Neural Networks (ANN), Activation Functions (ReLU, Sigmoid, Tanh), Keras, TensorFlow Basics

Dataset Used: MNIST Dataset / Sample Neural Network Dataset

Code Applied: Neural network creation, model compilation, training and prediction using Keras.

Day 14:

Topics Covered: Feature Engineering, Encoding Techniques, Missing Value Handling, Outlier Detection and Handling, Pipelines

Dataset Used: Customer Dataset, Employee Dataset, Sample ML Dataset

Code Applied: Label Encoding, One-Hot Encoding, Missing Value Imputation, Outlier Treatment, ML Pipeline Creation.

LABS AND METRICES:

Based on your Days 7–14 topics, you can send this in WhatsApp:

Day 7 Lab: Student Performance Prediction

- Dataset: Student Performance Dataset
- Algorithm: Supervised Learning Classification
- Metrics:
    - Accuracy
    - Precision

    - Recall
    - F1-Score
    - Confusion Matrix

Day 8 Lab: Salary/Student Score Prediction using Linear Regression

- Dataset: Student Study Hours Dataset, Salary Dataset
- Algorithm: Linear Regression
- Metrics:
    - R^2 Score
    - MAE (Mean Absolute Error)
    - RMSE (Root Mean Squared Error)

Day 9 Lab: Disease Prediction using Logistic Regression

- Dataset: Heart Disease Dataset, Diabetes Dataset
- Algorithm: Logistic Regression
- Metrics:
    - Accuracy
    - Precision
    - Recall
    - F1-Score
    - Confusion Matrix
    - ROC Curve
    - AUC Score

Day 10 Lab: Decision Tree Classification

- Dataset: Iris Dataset, Customer Classification Dataset
- Algorithm: Decision Tree
- Metrics:
    - Accuracy
    - Gini Impurity
    - Cross Validation Score

- Confusion Matrix

Day 11 Lab: Random Forest Classification

- Dataset: Iris Dataset, Customer Classification Dataset, Diabetes Dataset
- Algorithm: Random Forest
- Metrics:
  - Accuracy
  - OOB Score
  - Feature Importance
  - Test Score

Day 12 Lab: Customer Segmentation using K-Means Clustering

- Dataset: Iris Dataset, Sample Clustering Dataset
- Algorithm: K-Means Clustering
- Metrics:
  - WCSS (Within Cluster Sum of Squares)
  - Elbow Method
  - Silhouette Score

Day 13 Lab: Neural Network for Digit Classification

- Dataset: MNIST Dataset
- Algorithm: Artificial Neural Network (ANN) using Keras
- Metrics:
  - Accuracy
  - Loss
  - Validation Accuracy
  - Validation Loss

Day 14 Lab: Feature Engineering & Data Preprocessing

- Dataset: Customer Dataset, Employee Dataset
- Techniques:
  - Label Encoding

- One-Hot Encoding
- Missing Value Handling
- Outlier Treatment
- Pipeline Creation

- Metrics:
  - Missing Value Percentage
  - Feature Importance
  - Model Accuracy Before Preprocessing
  - Model Accuracy After Preprocessing

Summary of Metrics Learned:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix
- ROC Curve
- AUC Score
- R^2 Score
- MAE
- RMSE
- Gini Impurity
- Cross Validation Score
- OOB Score
- Feature Importance
- WCSS
- Silhouette Score
- Loss & Validation Loss